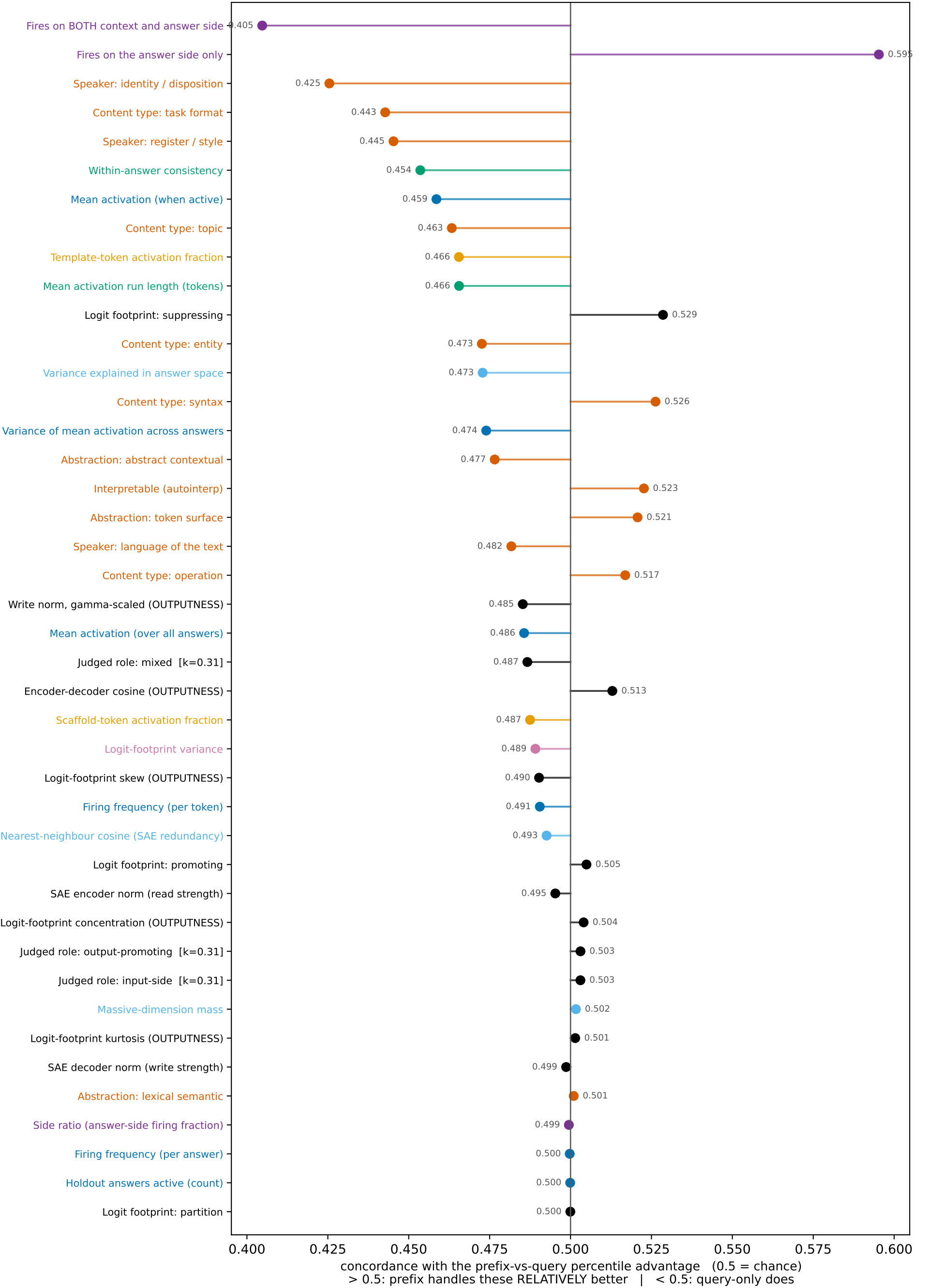


Which feature properties mark a PREFIX advantage over QUERY-ONLY?
outcome = $\text{pctile}(R^2_{\text{prefix}}) - \text{pctile}(R^2_{\text{query}})$, ranked within each arm · n = 113,288 features · all 42 predictors



● activity / firing frequency ● where in the text it fires ● output-side / logit footprint ● context- vs answer-side (token position)
● high- vs low-level proxy ● SAE geometry ● read- vs write-side (circuit role) ● LLM-judged property (binary)